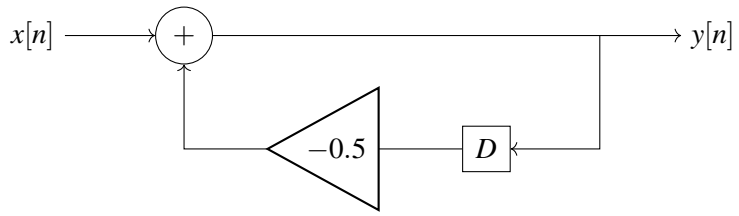
EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 10B

1. Frequency Response from a DAG Block Diagram [Adapted from EE 120]

Consider a DT LTI system with input $x[n]$ and output $y[n]$ that is implemented by the following block diagram:



(a) Find the LCCDE that describes this system.

(b) Find the frequency response $H(\omega)$ of this system.

UCB EE 66, Fall 2026, Discussion 10B, All Rights Reserved. This may not be publicly shared without explicit permission.

(c) Is this a high pass or low pass filter?

UCB EE 66, Fall 2026, Discussion 10B, All Rights Reserved. This may not be publicly shared without explicit permission.

2. 16A's Greatest Hits

Kevin is a DJ working on his newest audio filter, the Two Tap, which can be modelled as the following LCCDE:

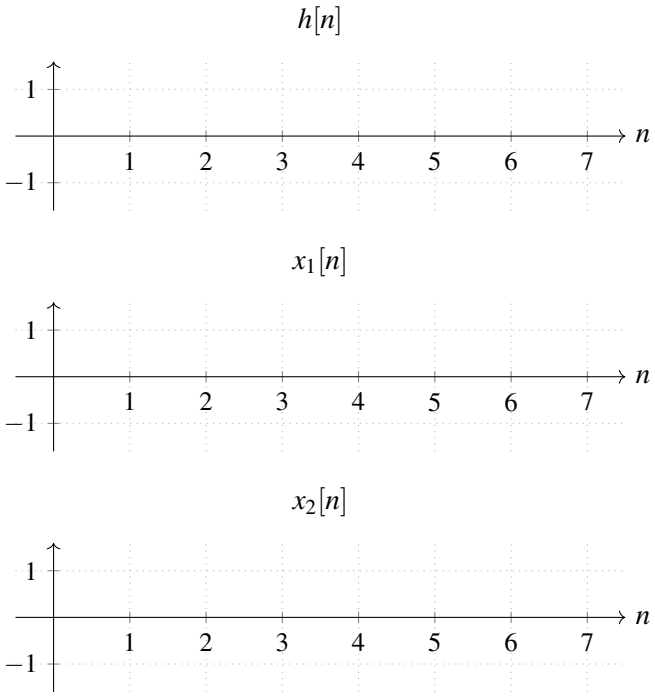
$$y[n] = x[n] + x[n-2].$$

Kevin would like to mix the following beat using his audio filter:

$$x[n] = (-1)^n + \cos\left(\frac{\pi n}{2}\right),$$

and would like to do so by passing it into the system.

(a) Kevin starts by trying to solve this by convolving the input signal. Sketch both components of the input signal below, then use your sketches and the system equation to find a closed-form expression for $y[n]$.



(b) Rival DJ Mallika sees Kevin running this computation, and would like to approach solving for the output to the Two Tap from a different perspective. To help Mallika, decompose $x[n]$ using a DTFS and identify the coefficients X_k by inspection.

(c) Suppose a general DT LTI system has frequency response $H(\omega)$, and the input $x[n]$ is periodic with DTFS coefficients X_k. Write a simple expression for the output $y[n]$.

(d) Using the expression from part (c), help Mallika find $y[n]$ for this system. First determine $H(\omega_k)$, then apply it to the DTFS coefficients from part (b). Verify Mallika's answer matches up with Kevin's.

UCB EE 66, Fall 2026, Discussion 10B, All Rights Reserved. This may not be publicly shared without explicit permission.